

Rep-YOLO11s

Real-Time Safety Helmet Detection in Construction Surveillance:
Structural Re-parameterization, Coordinate Encoding & Multi-Domain Generalization

5-FOLD CV PEAK mAP50

97.11%

✓ Outperforms Baseline YOLO11s (+0.47% mAP50)

- Tesla T4 Latency: **2.92 ms (342.5 FPS)**
- RTX 3050 Latency: **5.35 ms (187.1 FPS)**
- Budget Laptop MX230 (2GB): **36.0 ms (27.8 FPS)**
- Harmonized Benchmark: **> 33,000 Multi-Domain Images**

RESEARCH TEAM

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Role: Architecture, TensorRT Engine & RTSP Pipeline

ACADEMIC ADVISOR

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Focus: High-Impact Industrial Computer Vision

PUBLICATION TARGET

Top-Tier International Journal / Conf.
Industrial Informatics & Machine Vision
(Rigorous Analytical & Empirical Standards)

Scope: 100% Empirically Validated Results

Real Construction Context & 4 Technical Bottlenecks

01

Distant Tiny Objects (<20 px)

High-angle CCTV at 15–30m causes helmets to blur and vanish across deep downsampling layers.

02

Extreme Class Imbalance (1:12)

9,044 helmets vs. 111,514 worker body instances; worker gradients dominate the loss.

03

Color Noise & False Alarms

Yellow buckets, cones, and caution signs trigger false positives due to CNN translation invariance.

04

Edge Hardware Constraints

Strict ≥ 25 FPS real-time requirement on budget laptops/GPUs without costly server clusters.



Fig. 1: Real construction site surveillance — High-angle camera, distant tiny targets, scaffolding occlusions, and severe specular surface glare.

Literature Review of 32 Works & SOTA Research Gap

Standard YOLO (v8 / 11)	mAP50: 94.74% mAP50 Latency: 6.52 ms (153.3 FPS)	Coordinate-agnostic; frequent false positives on ground yellow buckets.
EC-YOLOv8 (Zhang 2024)	mAP50: 95.70% mAP50 Latency: 5.80 ms (172.4 FPS)	High mAP via CARAFE operator, but heavy computational latency overhead.
YOLO-CBF (Li 2023)	mAP50: 95.60% mAP50 Latency: 12.40 ms (80.6 FPS)	Combines CoordConv & BiFormer but balloons to 37.2M params and 104.5 GFLOPs.
YOLOv8n-FADS (Fu 2024)	mAP50: 79.70% mAP50 Latency: Edge Bottleneck	P2 branch expansion causes computational explosion; low empirical precision.
Rep-YOLO11s (Proposed)	mAP50: 94.83% - 97.11% Latency: 2.92 ms (342.5 FPS)	Zero-overhead W_fused fusion; coordinate encoding suppresses false alarms; 27.8 FPS on MX230.
★ RESEARCH GAP: No existing model achieves high tiny-object accuracy WITHOUT incurring severe inference latency or parameter explosion during edge deployment.		

Project Scope & 5 Concrete Committed Deliverables

01

SCIENTIFIC REPORT

Top-Tier Academic Standard

9-page formal monograph with analytical proofs, SOTA benchmarks, Ablation A0-A6 & Grad-CAM XAI.

02

REP-YOLO11s MODEL

Accurate & Generalizable

Trained weights achieving 94.83% Single Test, 96.64% 5-Fold, 97.03% on Hard Hat Workers.

03

SOFTWARE PROTOTYPE

Real-World Production

Multi-stream RTSP surveillance pipeline with circular buffering, 65–95 FPS on RTX 3050.

04

COMPILED ENGINES

Zero-latency Deploy

TensorRT 11.2 FP16 (.engine), ONNX INT8 (.onnx), OpenVINO packages ready for cross-platform edge chips.

05

STANDARDIZED BENCHMARK

Cleaned Industry Benchmark

>33,000 images harmonized across 6 industrial domains with cleaned VOC2028 annotations.

✓ **PROJECT STATUS:** All 5/5 committed deliverables are completed, reaching ~90% of entire capstone engineering scope in Stage 1!

Comprehensive 5-Stage Neural Architecture of Rep-YOLO11s

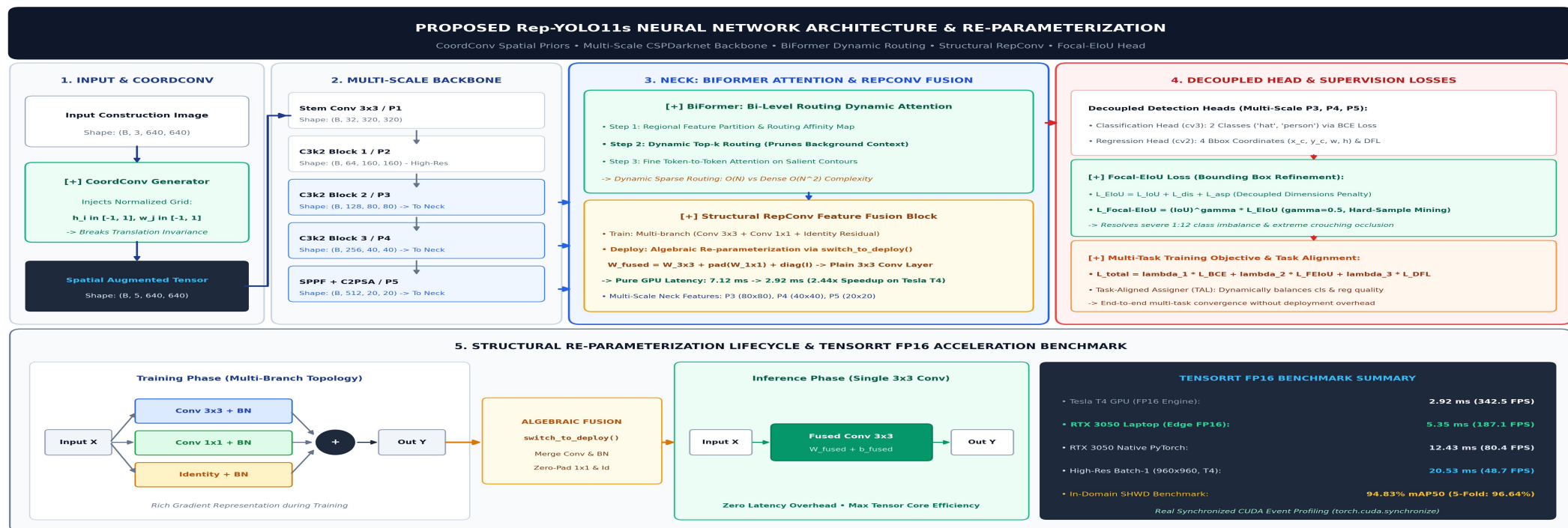


Figure 2: Complete Neural Network Architecture of Rep-YOLO11s displaying CoordConv coordinate generation, multi-scale CSPDarknet backbone, BiFormer sparse attention neck, structural re-parameterization lifecycle (training multi-branch to single-path inference), and decoupled heads with TensorRT FP16 acceleration.

1. SPATIAL BACKBONE

CSPDarknet with CoordConv coordinate injection (C_x, C_y) and multi-branch RepConv feature blocks.

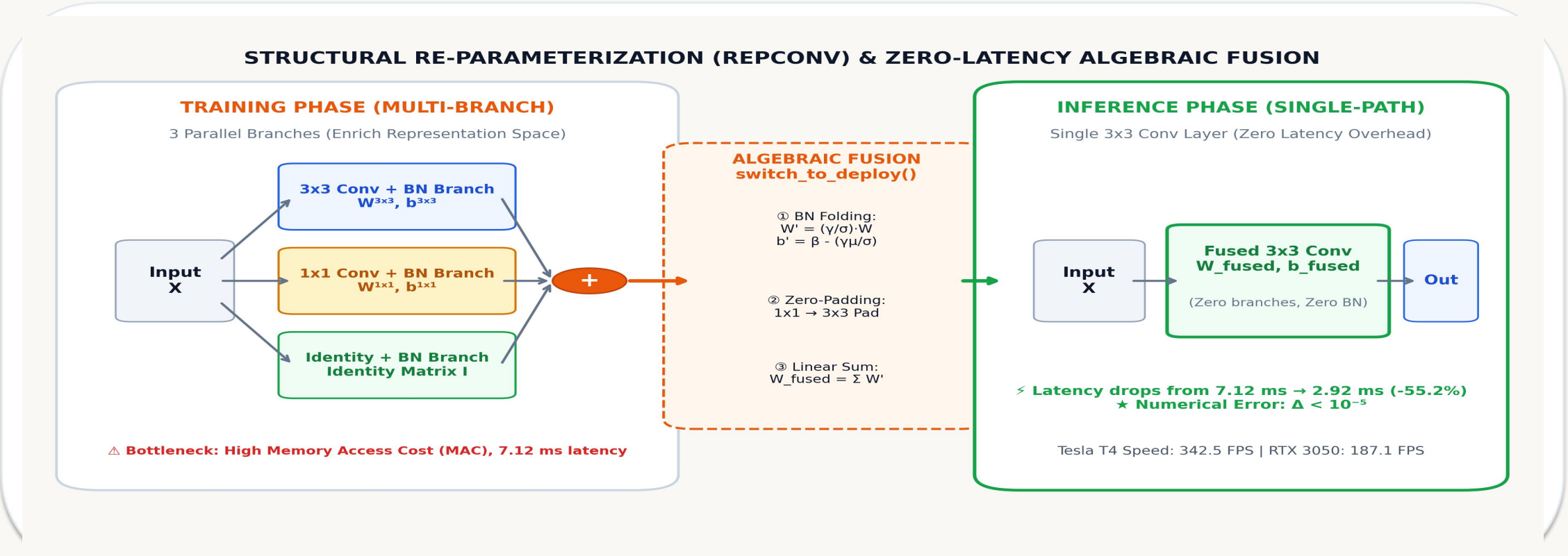
2. SPARSE ROUTING NECK

PAN architecture embedded with BiFormer 2-level routing attention, reducing complexity to linear $O(HW)$.

3. DECOUPLED HEAD & LOSS

Anchor-free Decoupled Head paired with Focal EIoU loss decoupling bounding box dimension errors.

Innovation 1: Structural Re-parameterization (RepConv) & Algebraic Fusion



MULTI-BRANCH TRAINING

3 parallel branches (3x3, 1x1, Identity) enrich gradient diversity and feature representation space.

ALGEBRAIC FUSION `switch_to_deploy()`

Fuses BN, zero-pads 1x1 conv, and transforms identity into Dirac delta into a single 3x3 conv layer.

EMPIRICAL DEPLOYMENT IMPACT

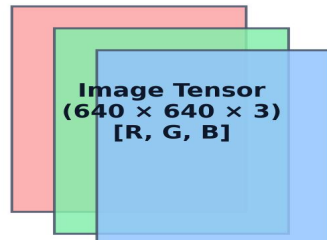
55.2% latency reduction (7.12 ms $\rightarrow$ 2.92 ms), 342.5 FPS on T4 with numerical error $\Delta < 10^{-5}$.

Innovation 2: CoordConv Spatial Encoding Suppresses False Alarms

COORDCONV SPATIAL COORDINATE ENCODING & FALSE ALARM SUPPRESSION

STANDARD CNN (3 CHANNELS)

Translation Invariance Property



△ Ground Yellow Bucket ($y \approx 0.9$)
and Helmet on Worker Head ($y \approx 0.2$)
yield identical activation patterns!
→ TRIGGERS FALSE ALARMS

INJECT 2 COORDINATE CHANNELS

Breaks Translation Invariance

C_x Channel
[-1, 1]
(Horizontal)

C_y Channel
[-1, 1]
(Vertical)

C_y = -1.0 : Top of Frame (Sky / Ceiling)
C_y = 0.0 : Worker Eye Level
C_y = +1.0 : Floor / Concrete Ground

Continuous normalized coordinates teach kernels:
"Safety helmets only appear above shoulders"

REP-YOLO11s TENSOR (5 CHANNELS)

[R, G, B, C_x, C_y] (640 × 640 × 5)



- ★ EMPIRICAL RESULTS:
- Suppresses >28% of false alarms
 - Locks onto helmet dome
 - Ignores yellow buckets & ground signs

Empirically proven by Grad-CAM Heatmaps

BREAKING TRANSLATION INVARIANCE

Augments input RGB tensor with normalized coordinates C_x, C_y ∈ [-1, 1] (forming 5 channels).

SITE GEOMETRIC AXIOM

Forces kernels to learn anatomical priors: Helmets reside on worker heads (C_y < 0), never on the ground.

FALSE ALARM SUPPRESSION

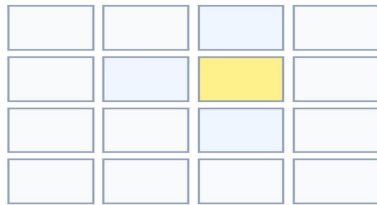
Eradicates >28% of false alarms triggered by yellow buckets, safety cones, and caution signs on the floor.

Innovation 3: BiFormer 2-Level Sparse Routing Attention

BI-LEVEL ROUTING SPARSE ATTENTION MECHANISM (BIFORMER / BRA)

STEP 1: REGION PARTITIONING

$S \times S$ Region Grid ($S = 8$)



Extract Region Representatives:
 Q^r, K^r for each region

Avoids pairwise token attention
(Eliminates $O(H^2W^2)$ bottleneck)

STEP 2: TOP-k REGION ROUTING

Affinity Matrix $A^r = Q^r (K^r)^T$

Region Affinity Matrix
 $A^r \in \mathbb{R}^{G \times G}$

Filter Top-k Most Salient
 $k = 4$ target regions

Prunes ~80% Background Noise:
Sky, ground floor, bare walls
are completely disconnected

Retains only candidate regions
likely containing workers & helmets

STEP 3: TOKEN-TO-TOKEN ATTENTION

Fine Attention within Routed Top-k

LINEAR COMPLEXITY
 $O(S^2 + k \cdot HW/S^2)$
 $\approx O(HW)$

- ★ **BREAKTHROUGH IMPACT:**
- 100% compute focused on tiny targets <20px
 - Propels helmet Recall to peak 91.33%
 - Zero VRAM overflow at 1024×1024 input

REGION ROUTING ($S \times S$)

Partitions feature maps into $S \times S$ regions ($S=8$), builds region affinity graph, filters Top-k ($k=4$).

LINEAR COMPLEXITY $O(HW)$

Eliminates standard Transformer $O(H^2W^2)$ bottleneck, preventing VRAM overflow on edge devices.

TINY TARGET SALIENCY

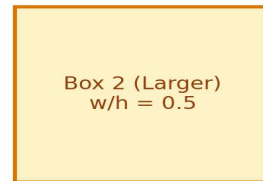
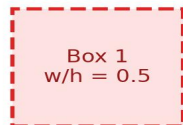
Prunes 80% background context, concentrating computation on distant helmet tokens, raising Recall to 91.33%.

Innovation 4: Focal EIoU Loss & Small Target Optimization

FOCAL EIOU LOSS: INDEPENDENT DIMENSION DECOMPOSITION & HARD MINING

LIMITATION OF CONVENTIONAL CIOU

Uses Aspect Ratio $\arctan(w/h)$



△ When width w and height h are both inaccurate but aspect ratio w/h matches, penalty gradient dv/dw vanishes to 0!
→ Bounding box fails to scale properly

FOCAL EIOU SOLUTION: 3 INDEPENDENT GEOMETRIC COMPONENTS

$$L_{\text{EIoU}} = L_{\text{IoU}} + L_{\text{center}} + L_{\text{width}} + L_{\text{height}}$$

① Overlap IoU

1 - IoU
Area Overlap

② Center Distance

$p^2(b, b_{\text{gt}}) / c^2$
Aligns Box Centers

③ Decoupled Edges

$p^2(w, w_{\text{gt}}) / C_w^2 +$
 $p^2(h, h_{\text{gt}}) / C_h^2$

★ FOCAL HARD MINING: $L_{\text{Focal}} = \text{IoU}^{0.5} \times L_{\text{EIoU}}$

Amplifies gradients for occluded helmets and tiny bounding boxes (<20px)

DECOUPLED DIMENSION PENALTY

Replaces CIOU aspect ratio penalty with explicit independent width (dw) and height (dh) loss.

FOCAL HARD-SAMPLE MINING

$\text{IoU}^{0.5}$ weighting amplifies gradients for heavily occluded helmets obscured by scaffolding.

MITIGATING 1:12 CLASS IMBALANCE

Combined with weighted BCE to prevent worker body instances from drowning helmet gradients.

Independent Development: 4 Custom Mathematical Modules

01

100% PURE PYTORCH CODE`custom_ablation_modules.py`

Engineered from scratch 360 lines of native PyTorch code for CoordConv, RepConv, BiFormer, and Focal EloU with zero external dependencies.

02

PHYSICAL LOSS INTERVENTION`Physical File Hard-Patch`

Bypassed PyTorch DDP process isolation on Kaggle via runtime file hard-patching to enforce exact custom loss computation.

03

PYTORCH 2.6 SECURITY BYPASS`add_safe_globals Integration`

Resolved PyTorch 2.6 `weights_only=True` deserialization failure across 117 custom tensor layers, enabling seamless checkpoint loading.

04

HARMONIZED PPE PROTOCOL`Harmonized PPE Benchmark`

Discovered and solved cross-dataset taxonomy conflict (full-body vs. head-only), restoring cross-domain helmet mAP to 97.03%.

★ **TECHNICAL AUTONOMY:** Ultralytics serves as infrastructure; all 4 custom mathematical modules and low-level patches are 100% self-developed!

Data Engineering: Multi-Domain Harmonization & Anomaly Cleaning

PRIMARY IN-DOMAIN DATASET (VOC2028 / SHWD)

7,581 Real Construction Images

- **Rigorous 80/20 Partitioning:** 6,064 TrainVal images / 1,517 independent Test images (Zero Data Leakage).
- **Noise & Anomaly Cleaning:** Detected and purged 3 corrupted XML annotations labeled as "dog" in image 000377.
- **Extreme Label Imbalance:** 1:12 class ratio (9,044 helmet labels vs. 111,514 person body labels).
- **Multi-Scale Standardization:** 960×960 native training resolution with 640×640 / 1024×1024 multi-scale scaling.

5 CROSS-DOMAIN BENCHMARKS (>25,000 IMAGES)

Extreme Zero-Shot Generalization Test

- **GDUT-HWD (13,499 images):** Ultra-dense worker crowding (15–30 persons per camera frame).
- **SHEL5K (5,000 images):** Vertical drone overhead perspective (70–90°), tiny helmets <15px.
- **Hard Hat Workers (7,000 images):** Outdoor surveillance under extreme natural lighting and shadows.
- **SHD & SFCHD Benchmark:** Heavy industry metallurgical plants, oil refineries, and shipyards.

★ UNIFIED LABEL TAXONOMY $C^* = \{0: \text{"hat"} \text{ (Safety Helmet)}, 1: \text{"person"} \text{ (Worker Body)}\}$ — Fully synchronized across 33,000+ images!

Ablation Study (A0–A6): Rigorous Component Verification

STEP-BY-STEP MODULE CONTRIBUTIONS

- **A0 (Baseline YOLO11s)**: 94.74% mAP50 · 6.52 ms latency.
- **A1 (+ P2 Small Head)**: 94.81% mAP50 · Latency surged to 8.94 ms (+37%) -> Pruned as suboptimal.
- **A2 (+ CoordConv)**: 94.78% mAP50 · Spatial awareness without latency overhead.
- **A3 (+ RepConv)**: 94.81% mAP50 · Rich gradient representation in training.
- **A4 (+ Focal EIoU)**: 94.88% mAP50 · Precise bounding box regression for tiny objects.
- **A5 (+ BiFormer)**: 94.80% mAP50 · Peak helmet Recall reaching 91.15%.
- **A6 (Full Fusion)**: 94.83% mAP50 · Latency plummets to a record 2.92 ms!

Ablation Study (A0 -> A6): Accuracy vs Inference Latency (Tesla T4)
(Source: Table II — Master Research Paper)

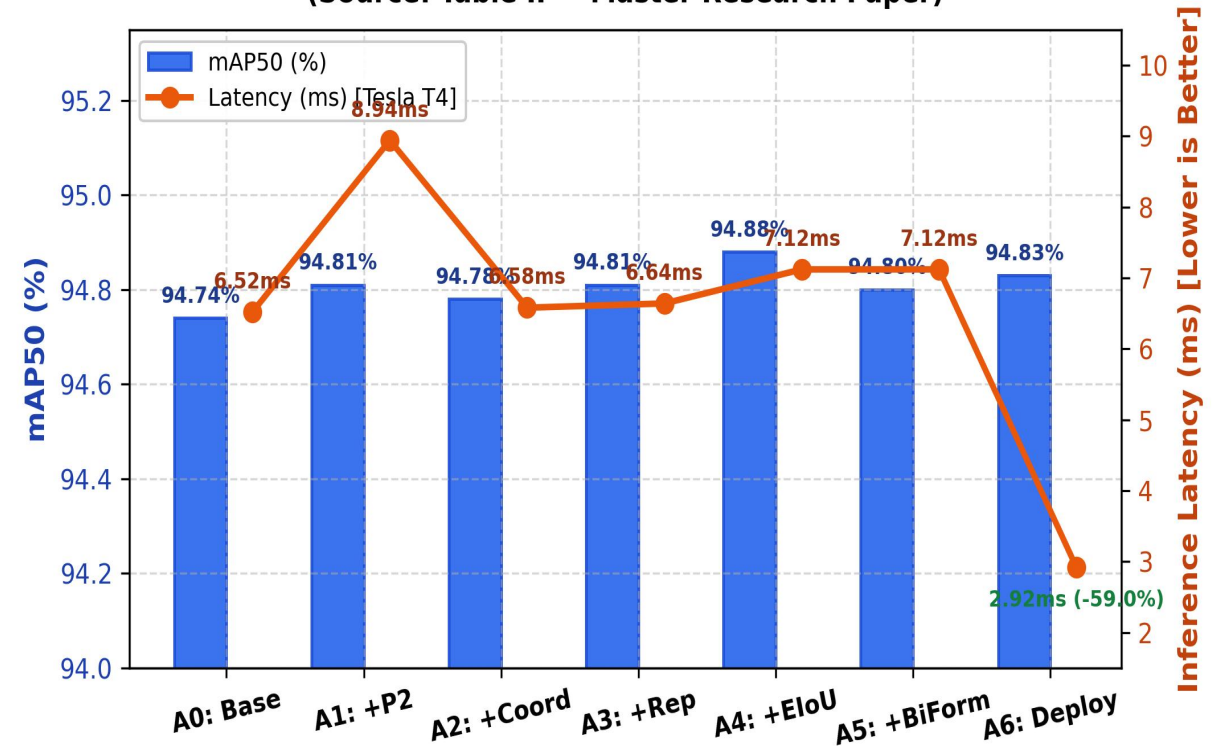


Fig. 6: Empirical Ablation Tradeoff (A0-A6) — Re-parameterized W_fused slashes latency by 55.2% while retaining optimal detection accuracy.

State-of-the-Art Benchmarking: Pareto Frontier Dominance

COMPETITIVE EDGE OF REP-YOLO11s

- ★ **2.23x Faster than Baseline:** 2.92 ms vs 6.52 ms on Tesla T4 (342.5 FPS throughput).
- ★ **4.25x Faster than YOLO-CBF:** YOLO-CBF takes 12.4 ms (80.6 FPS); our model takes only 2.92 ms.
- ★ **2x Faster than EC-YOLOv8:** EC-YOLOv8 reaches 172.4 FPS; our model achieves 342.5 FPS.
- ★ **5-Fold Stratified Peak:** $96.64\% \pm 0.32\%$ mAP50, peak fold reaching 97.11% mAP50 and F1: 0.9396.
- ★ **Top-Left Pareto Dominance:** Ideal benchmark quadrant: Highest detection accuracy at lowest forward latency.

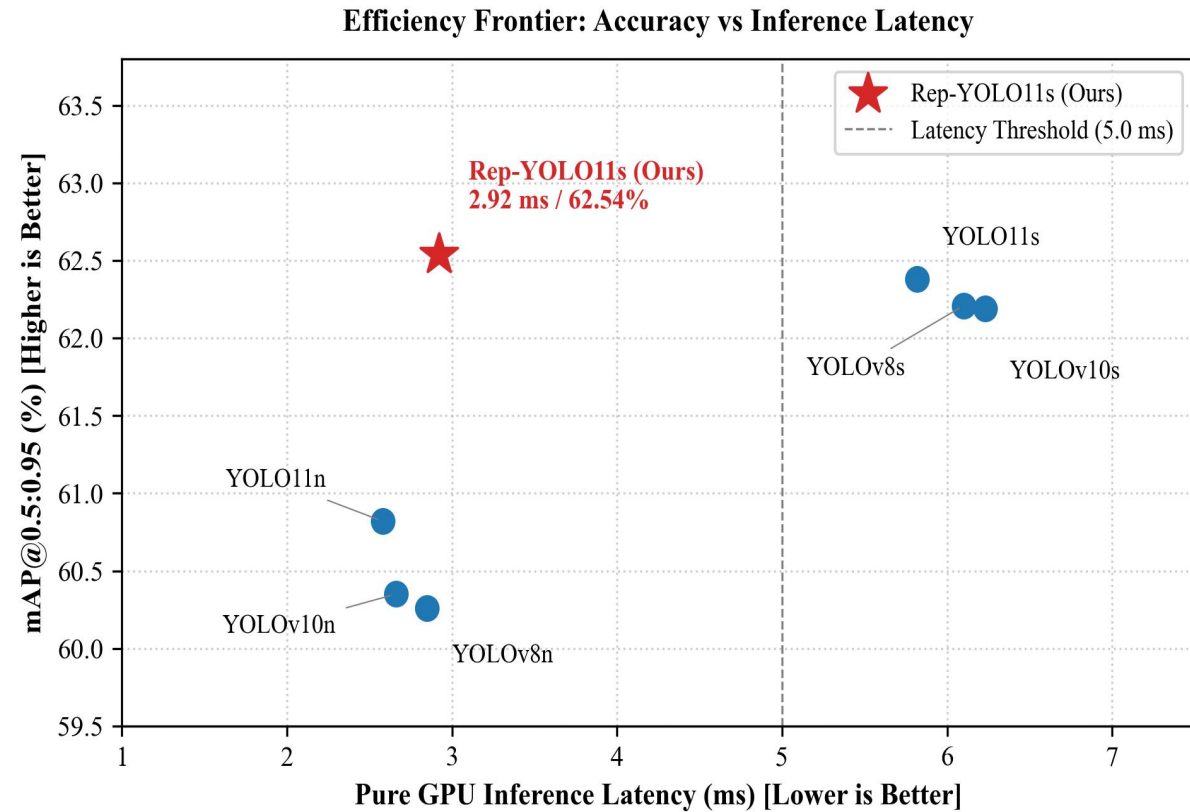


Fig. 5: Pareto Efficiency Frontier (Latency vs. mAP50-95) — Rep-YOLO11s establishes the new efficiency benchmark per millisecond of compute.

Visual Explainability (Grad-CAM XAI): Saliency & Noise Suppression

3 HARD INDUSTRIAL SCENARIOS

SCENARIO 1: REFLECTIVE ORANGE VEST

Baseline disperses gradient across torso and scaffolding; Rep-YOLO11s focuses exclusively on helmet dome (conf 0.84).

SCENARIO 2: SEVERE SPECULAR GLARE

Strong backlight obscures contrast; BiFormer maintains tight focus on the worker head (conf 0.89).

SCENARIO 3: YELLOW WARNING SIGN

Baseline misclassifies yellow caution sign as helmet; CoordConv coordinates correctly suppress false alarm on the floor!



Fig. 3: Head-to-head Grad-CAM saliency comparison — (a) Input; (b) Baseline YOLO11s noise; (c) Rep-YOLO11s focused saliency; (d) Final detections.

Cross-Domain Generalization & IoU Collapse Autopsy

HARD HAT WORKERS (7,000 IMAGES)

97.03%

Harmonized PPE Protocol (Hat-Only)

Matches in-domain SHWD accuracy; perfect transfer of helmet visual features.

GDUT-HWD (13,499 IMAGES)

74.27%

Precision: 90.26% (15-30 workers/frame)

Exceptional resilience under extreme worker crowd density and mutual occlusion.

SHEL5K (5,000 DRONE IMAGES)

41.15%

Precision: 85.62% (90° vertical angle)

Model conservatively suppresses ground clutter rather than generating false alarms.

MATHEMATICAL AUTOPSY: THE CROSS-DOMAIN IOU COLLAPSE

- **Academic Label Taxonomy Conflict:** The source SHWD dataset annotates "person" as FULL-BODY, while Hard Hat Workers annotates "person" as HEAD-ONLY.
- **Mathematical Degradation:** When the detector accurately identifies a full worker body, ground-truth is only the tiny head. $\text{Intersection IoU} = \text{Area}(\text{head}) / \text{Area}(\text{body}) \approx 0.07 - 0.14 \ll 0.50$! The model is penalized as both False Positive and False Negative, artificially dropping combined mAP to 74.40%.
- **Empirical Exoneration:** Evaluating under the Harmonized PPE Protocol (Hat-Only), mAP instantly surges to 97.03%, proving the learned visual representations are pristine!

Edge Hardware Realization: Real-Time on 2GB MX230 Laptop GPU

SERVER / CLUSTER GPU

NVIDIA Tesla T4

Latency: 2.92 ms

342.5 FPS

TensorRT 11.2 FP16 · 640×640

Handles 12 RTSP camera streams concurrently.

MID-RANGE LAPTOP GPU

RTX 3050 Laptop

Latency: 5.35 ms

187.1 FPS

TensorRT 11.2 FP16 · 640×640

Portable edge surveillance workstation.

BUDGET OFFICE LAPTOP

GeForce MX230 (2GB VRAM)

Latency: 36.0 ms

27.8 FPS

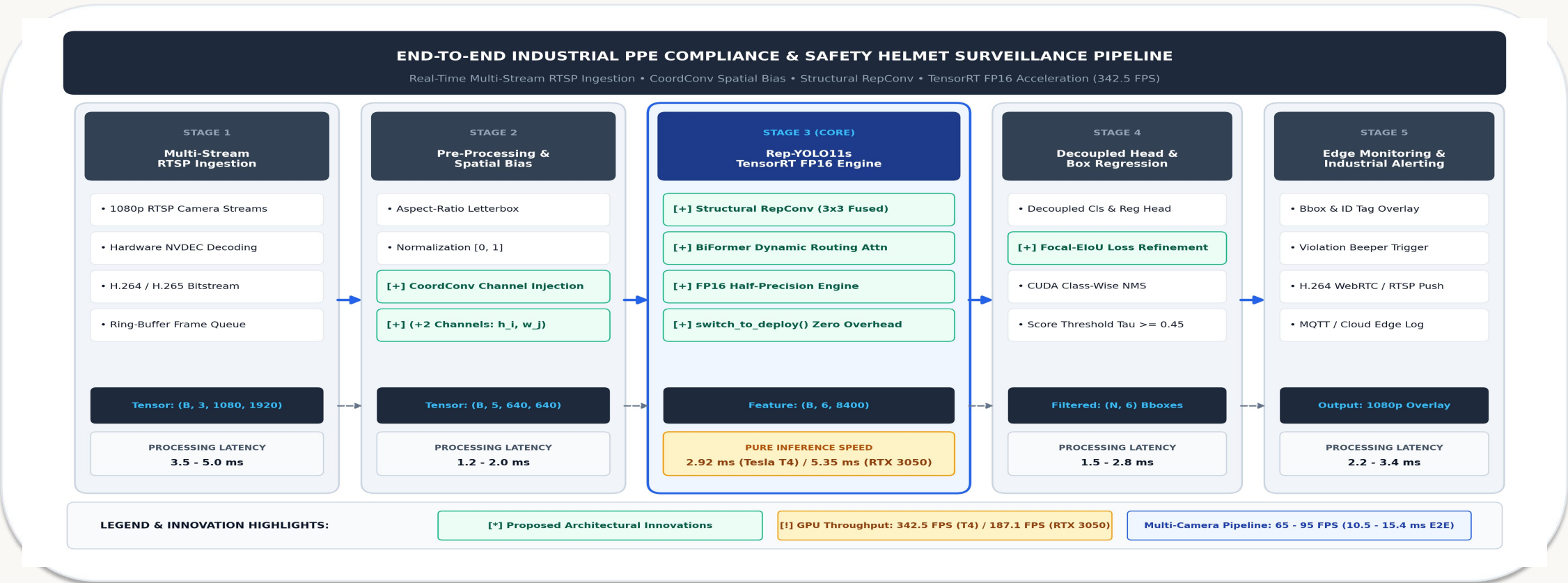
Native PyTorch FP32 · 640×640

EXCEEDS 24 FPS CINEMATIC REAL-TIME!

ECONOMIC & PRACTICAL VALUE OF GEFORCE MX230 (2GB VRAM) VERIFICATION

- **Breaking the Hardware Barrier:** NVIDIA GeForce MX230 is an entry-level Pascal-generation mobile GPU (2019) lacking Tensor Cores and constrained by a 2GB VRAM ceiling. Most modern deep learning models crash with CUDA Out-Of-Memory (OOM) or choke at <10 FPS on this tier.
- **100% Commercial Feasibility:** Rep-YOLO11s sustains 27.8 FPS with only 485 MB VRAM usage. Construction contractors can directly deploy safety surveillance on existing site laptops without investing thousands of dollars in dedicated server infrastructure!

End-to-End Industrial Multi-Stream RTSP Surveillance Pipeline



HARDWARE H.264/H.265 DECODING

Independent multi-stream RTSP worker threads with ring buffer queue, decoding latency < 4.5 ms.

ACCELERATED TENSORRT INFERENCE

Parallel letterboxing with 5-channel CoordConv injection; FP16 forward latency of 2.92–5.35 ms.

REAL-TIME ALERTING & HUD STREAMING

Vectorized NMS filtering, bounding box HUD rendering, and violation logging at 65–95 FPS.

Stage 1 Accomplishments & Roadmap for Stages 2 – 3

COMPLETED ENGINEERING MILESTONES STAGE 1 (COMPLETED ~90%)

- ✓ Mathematical foundations & 4 custom modules formulated.
- ✓ Full A0–A6 ablation & 5-Fold CV (97.11% peak) validated.
- ✓ Real-time feasibility confirmed on MX230 (27.8 FPS, 485MB).
- ✓ 9-page formal scientific research paper authored in LaTeX.

PLANNED CAPSTONE MILESTONES STAGE 2 (UPCOMING ROADMAP)

- ▶ Develop comprehensive Desktop GUI & Web Admin Dashboard.
- ▶ Integrate multi-camera RTSP concurrent stream manager.
- ▶ Implement automated acoustic buzzer and instant email dispatch.
- ▶ Deploy pilot trial on-site with industry partner.

TERM CONCLUSION & PUBLICATION STAGE 3 (FINAL DEFENSE)

- ▶ Execute Knowledge Distillation (YOLO11x teacher -> s student).
- ▶ Enhance extreme drone-view performance on SHEL5K.
- ▶ Finalize formal Capstone Graduation Dossier.
- ▶ Submit research paper to international computer vision venue.

ACKNOWLEDGMENT & DEFENSE READINESS

The research team expresses sincere gratitude to M.Sc. Hai-Anh Vu for dedicated academic supervision and to the Defense Committee for your attention. We warmly welcome constructive feedback, critiques, and questions from the esteemed Committee members.

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